Case Study No.: 5 **Name:** Rohan Laharwani **PRN:** 24070521055

1. Experiment Title:

Create an array of numbers and find the maximum and minimum

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Case Study 5.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Array Min & Max Finder</title>
    <link rel="stylesheet" href="Case Study 5.css">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
    <div class="container">
        <h2>Array Min & Max Finder</h2>

        <div class="input-card">
            <div class="input-group">
                <input type="number" id="numberInput" placeholder="Enter a number">
                <button class="btn btn-primary" onclick="addNumber()">Add Number</button>
                <button class="btn btn-secondary" onclick="addRandomNumber()">Add Random</button>
                <button class="btn btn-danger" onclick="resetArray()">Reset</button>
            </div>
        </div>

        <div class="display-card">
            <h3>Current Array</h3>
            <div id="arrayDisplay" class="array-container">
                <span class="empty-msg">Array is currently empty</span>
            </div>
        </div>

        <div class="stats-grid">
            <div class="stat-card max-card">
                <div class="stat-label">Maximum Value</div>
                <div id="maxValue" class="stat-value">-</div>
            </div>
            <div class="stat-card min-card">
                <div class="stat-label">Minimum Value</div>
                <div id="minValue" class="stat-value">-</div>
            </div>
            <div class="stat-card count-card">
                <div class="stat-label">Total Elements</div>
```

```
                <div id="totalCount" class="stat-value">0</div>
            </div>
        </div>

        <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-
top: 1px solid lightgray; padding-top: 15px;">
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Case Study 5.js"></script>
</body>
</html>
```

Case Study 5.css

```
* {
    box-sizing: border-box;
}

body {
    background-color: whitesmoke;
    color: dimgray;
    font-family: sans-serif;
    margin: 0;
    padding: 40px 20px;
    display: flex;
    justify-content: center;
}

.container {
    width: 100%;
    max-width: 700px;
    background-color: white;
    padding: 30px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 15px lightgray;
}

h2 {
    color: darkblue;
    text-align: center;
    margin-top: 0;
    margin-bottom: 24px;
    font-size: 24px;
}

.input-card, .display-card {
    background-color: whitesmoke;
    padding: 20px;
    border: 1px solid lightgray;
    border-radius: 8px;
    margin-bottom: 20px;
}

.input-group {
    display: flex;
    gap: 12px;
    flex-wrap: wrap;
}

input {
    flex: 1 1 140px;
    min-width: 0;
    padding: 10px 14px;
    border: 1px solid lightgray;
    border-radius: 6px;
```

```
    font-size: 14px;
    outline: none;
    transition: border-color 0.2s;
}

input:focus {
    border-color: royalblue;
}

.btn {
    padding: 10px 18px;
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 14px;
    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s;
}

.btn:hover {
    background-color: blue;
}

.btn-secondary {
    background-color: teal;
}

.btn-secondary:hover {
    background-color: darkcyan;
}

.btn-danger {
    background-color: crimson;
}

.btn-danger:hover {
    background-color: darkred;
}

.array-container {
    display: flex;
    flex-wrap: wrap;
    gap: 10px;
    background-color: white;
    padding: 14px;
    border: 1px solid lightgray;
    border-radius: 6px;
    min-height: 48px;
    align-items: center;
}

.number-chip {
    padding: 6px 14px;
    background-color: whitesmoke;
    border: 1px solid lightgray;
    border-radius: 16px;
    font-size: 14px;
    font-weight: 600;
    color: black;
}

.stats-grid {
    display: flex;
    gap: 16px;
    justify-content: space-between;
}
```

```
.stat-card {
    flex: 1;
    background-color: whitesmoke;
    border: 1px solid lightgray;
    border-radius: 8px;
    padding: 18px;
    text-align: center;
}

.stat-label {
    font-size: 13px;
    color: dimgray;
    margin-bottom: 6px;
    font-weight: 600;
    text-transform: uppercase;
}

.stat-value {
    font-size: 24px;
    font-weight: 700;
    color: darkblue;
}
```

Case Study 5.js

```
let numbers = [];

function addNumber() {
    let val = parseFloat(document.getElementById("numberInput").value);

    if (isNaN(val)) {
        alert("Please enter a valid number");
        return;
    }

    let item = {
        id: numbers.length + 1,
        value: val
    };

    numbers.push(item);
    displayData();

    document.getElementById("numberInput").value = "";
}

function addRandomNumber() {
    let randomVal = Math.floor(Math.random() * 100) + 1;
    let item = {
        id: numbers.length + 1,
        value: randomVal
    };

    numbers.push(item);
    displayData();
}

function resetArray() {
    numbers = [];
    displayData();
}

function displayData() {
```

```
    let display = document.getElementById("arrayDisplay");
    display.innerHTML = "";

    numbers.forEach(function(item) {
        let chip = document.createElement("span");
        chip.className = "number-chip";
        chip.textContent = item.value;
        display.appendChild(chip);
    });

    document.getElementById("totalCount").textContent = numbers.length;

    if (numbers.length === 0) {
        display.innerHTML = '<span class="empty-msg">Array is currently empty</span>';
        document.getElementById("maxValue").textContent = "-";
        document.getElementById("minValue").textContent = "-";
        return;
    }

    let values = numbers.map(function(item) {
        return item.value;
    });

    let maxVal = values.reduce(function(max, val) {
        return val > max ? val : max;
    }, values[0]);

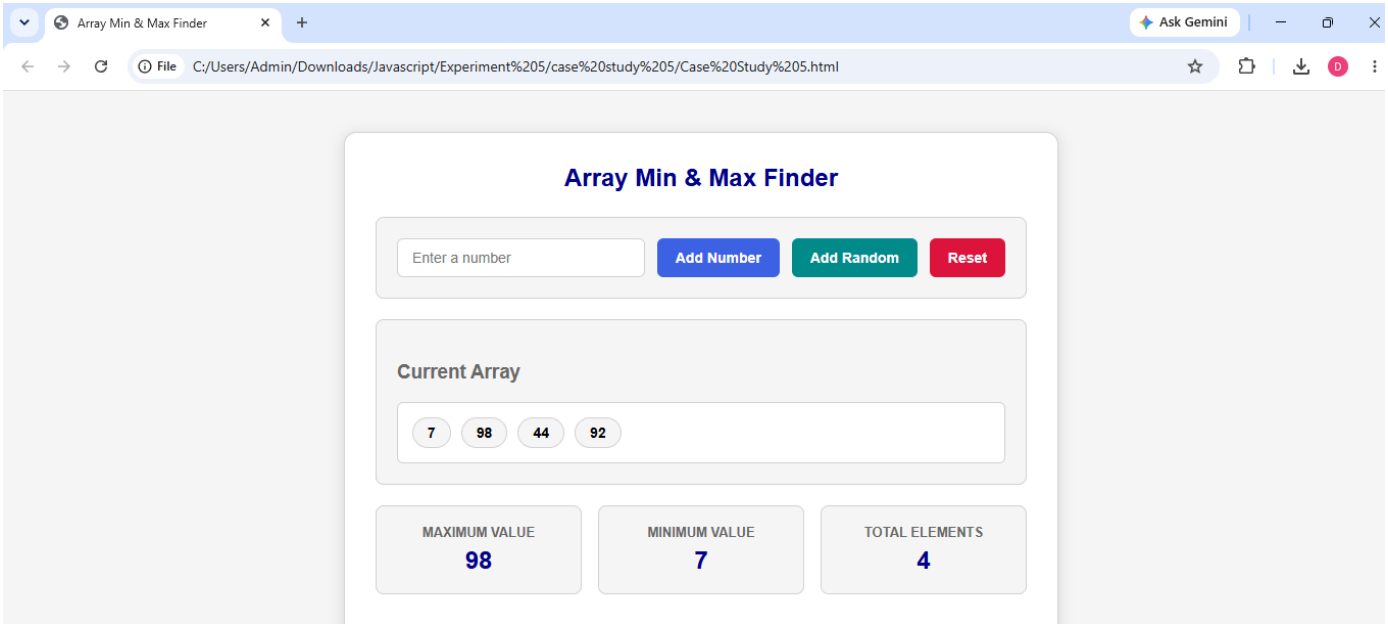
    let minVal = values.reduce(function(min, val) {
        return val < min ? val : min;
    }, values[0]);

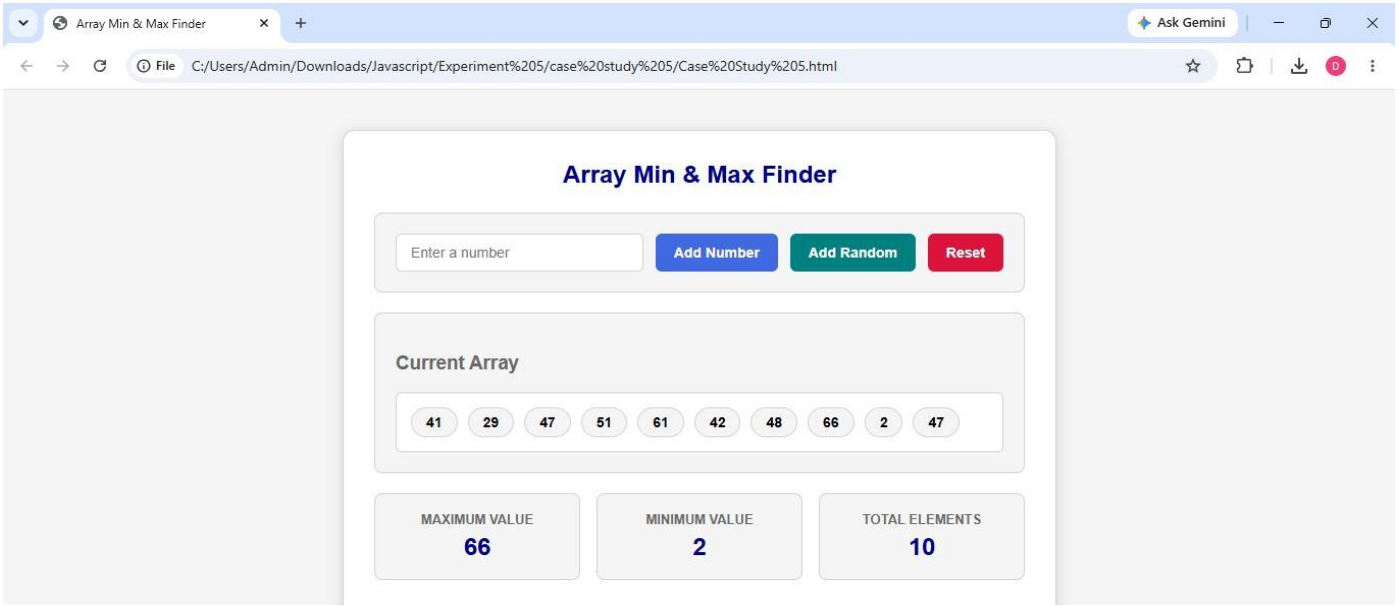
    document.getElementById("maxValue").textContent = maxVal;
    document.getElementById("minValue").textContent = minVal;
}

document.getElementById("numberInput").addEventListener("keypress", function(event) {
    if (event.key === "Enter") {
        addNumber();
    }
});

displayData();
```

4. Output:


Array Min & Max Finder
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/Experiment%205/case%20study%205/Case%20Study%205.html
Array Min & Max Finder
Enter a number
Add Number
Add Random
Reset
Current Array
7 98 44 92
MAXIMUM VALUE
98
MINIMUM VALUE
7
TOTAL ELEMENTS
4


Array Min & Max Finder
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/Experiment%205/case%20study%205/Case%20Study%205.html
Array Min & Max Finder
Enter a number
Add Number
Add Random
Reset
Current Array
41 29 47 51 61 42 48 66 2 47
MAXIMUM VALUE
66
MINIMUM VALUE
2
TOTAL ELEMENTS
10

5. Result/Conclusion:

A web application was developed to dynamically build an array of numbers, either through user input or randomly generated values, and display it on the page. The array was mapped to an array of values using map(), and reduce() was used twice to compute the maximum and minimum values by comparing each element against an accumulator. The total element count and array contents were updated dynamically on every change. This case study reinforced dynamic array construction, DOM manipulation, and the use of the map() and reduce() array methods in JavaScript.